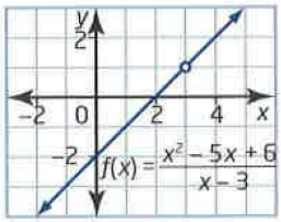
2.1/2.2 Multiplying and Dividing Rational Expressions – Worksheet

MCR3U

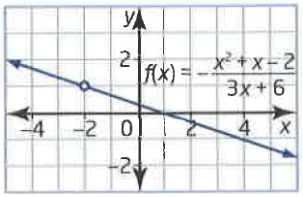
Jensen

1) State the restrictions for each function.

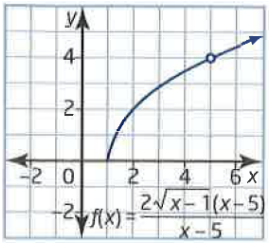
a)



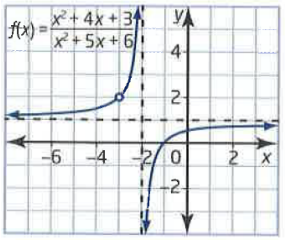
b)



c)



d)



2) Simplify each expression and state all restrictions on x.

a) $\frac{x-8}{x^2-13x+40}$

b) $\frac{x^2-3x-18}{x^2+x-42}$

c) $\frac{x+8}{x^2+6x-16}$

3) Simplify and state the restrictions on the variables.

a) $\frac{14y}{11x} \times \frac{121y}{7x}$

b) $\frac{15b^3}{4b} \times \frac{20b}{30b^2}$

c) $\frac{5x}{9y} \div \frac{5x}{18y^2}$

d) $\frac{26ab}{4a} \div \frac{39a^4b^3}{12b^4}$

4) Simplify and state the restrictions on the variable.

a) $\frac{25}{x+10} \times \frac{x+10}{5}$

b) $\frac{x+5}{x-3} \times \frac{x-3}{x+7}$

c) $\frac{x+1}{x} \div \frac{x+1}{2x}$

d) $\frac{x+12}{x+10} \div \frac{x+12}{x-5}$

5) Simplify and state the restrictions on the variable.

a) $\frac{3x^2}{12x^2+18x} \times \frac{4x+6}{3x+30}$

b) $\frac{4x+24}{x^2+8x} \times \frac{12x^2}{3x+18}$

c) $\frac{x^2+10x+21}{x+3} \times \frac{x+2}{x^2+9x+14}$

d) $\frac{x^2+2x-15}{x^2-9x+18} \times \frac{x-6}{x+5}$

6) Simplify and state the restrictions on the variable

a) $\frac{x^2+15x}{4x+24} \div \frac{3x}{3x+18}$

b) $\frac{6x}{8x-72} \div \frac{9x}{2x-18}$

c) $\frac{x^2+15x+26}{6x^2} \div \frac{x^2-3x-10}{30x^3}$

d) $\frac{x^2+11x+24}{x^2+2x-3} \div \frac{x-8}{x-1}$

7) Simplify and state the restrictions on the variable

a) $\frac{a^2-25}{a+2} \cdot \frac{a^2-4}{a^2-7a+10}$

b) $\frac{y^2-4y-21}{3y^2+6y} \cdot \frac{y^2+8y}{y^2+11y+24}$

c) $\frac{p^2-2p+1}{p+1} \div \frac{p^2-1}{p+1}$

d) $\frac{x^2+6x-27}{x^2+11x+18} \div \frac{x-3}{x^2+x-2}$

Answers

1) a) $x \neq 3$ **b)** $x \neq -2$ **c)** $x \geq 1,\ x \neq 5$ **d)** $x \neq -3,\ x \neq -2$

2) a) $\frac{1}{x-5}, x \neq 5,\ x \neq 8$ **b)** $\frac{x+3}{x+7}, x \neq -7,\ x \neq 6$ **c)** $\frac{1}{x-2}, x \neq -8,\ x \neq 2$

3) a) $\frac{22y^2}{x^2},\ x \neq 0$ **b)** $\frac{5b}{2},\ b \neq 0$ **c)** $2y,\ x \neq 0,\ y \neq 0$ **d)** $\frac{2b^2}{a^4},\ a \neq 0,\ b \neq 0$

4) a) $5, x \neq -10$ **b)** $\frac{x+5}{x+7},\ x \neq -7,\ x \neq 3$ **c)** $2,\ x \neq -1, x \neq 0$ **d)** $\frac{x-5}{x+10},\ x \neq -12, x \neq -10, x \neq 5$

5) a) $\frac{x}{3(x+10)}, x \neq -10, -\frac{3}{2}, 0$ **b)** $\frac{16x}{x+8}, x \neq -8, -6, 0$ **c)** $1, x \neq -7, -3, -2$ **d)** $1, x \neq -5,\ 3,\ 6$

6) a) $\frac{x+15}{4},\ x \neq -6, 0$ **b)** $\frac{1}{6},\ x \neq 0,\ 9$ **c)** $\frac{5x(x+13)}{x-5},\ x \neq -2, 0, 5$ **d)** $\frac{x+8}{x-8}, x \neq -3, 1, 8$

7) a) $a + 5, a \neq 2, -2, 5$ **b)** $\frac{y-7}{3(y+2)}, y \neq -8, -3, -2, 0$ **c)** $\frac{p-1}{p+1}, p \neq -1, 1$ **d)** $x - 1, x \neq -9, -2, 1, 3$